



ICE-4101

Artificial Intelligence and
Neural Networks

Lecture-5

Agent, Rational Agent, PEAS, Agent Terminology

Prepared By

Prof. Dr. Md. Ashikur Rahman Khan



What will we learn here!

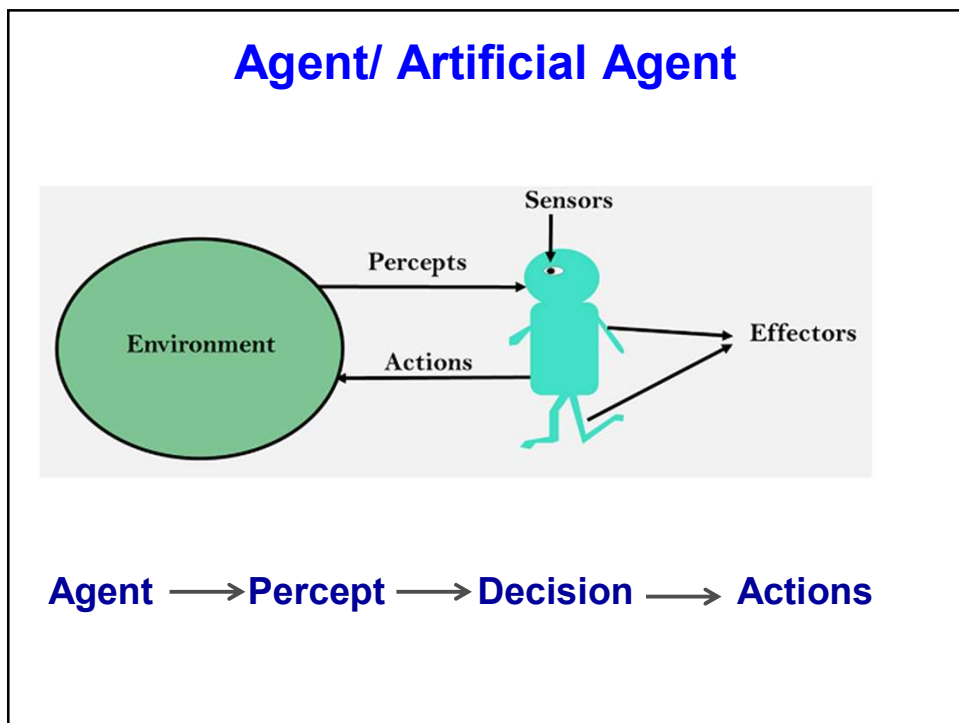
- ✓ **Agent**
 - ✓ **Sensors, Actuators**
 - ✓ **Types of Sensors**
- ✓ **Rational Agent**
- ✓ **PEAS**

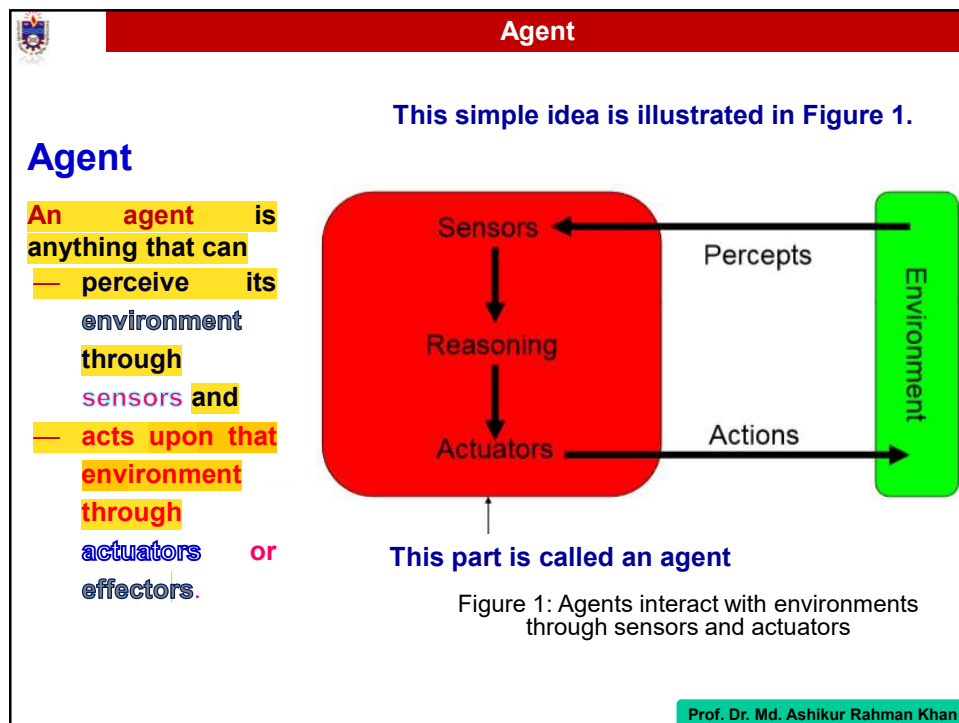
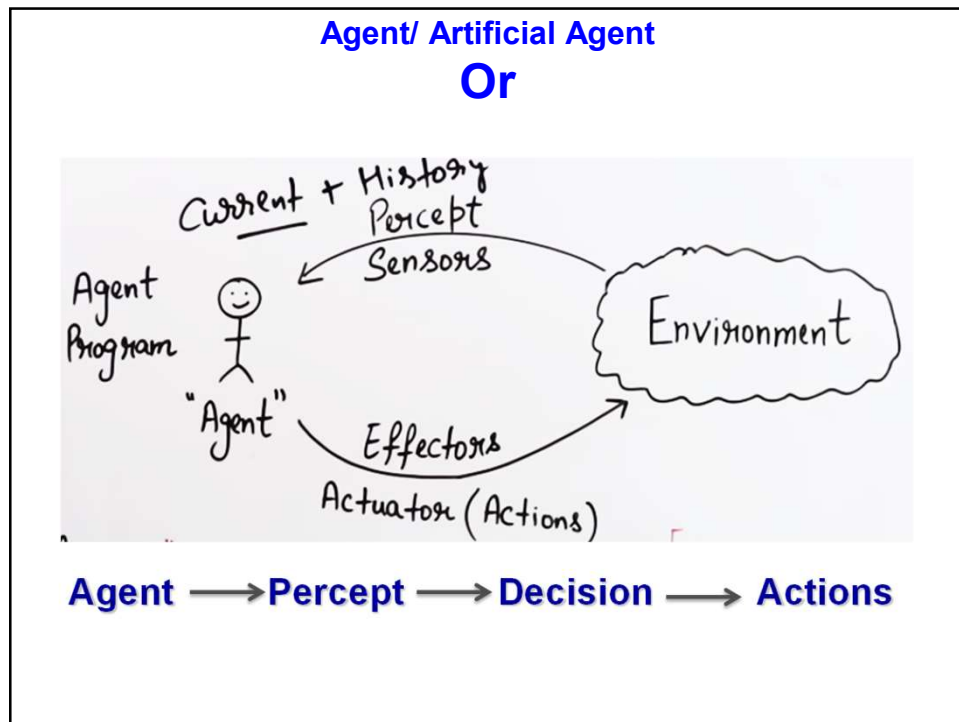
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Agent

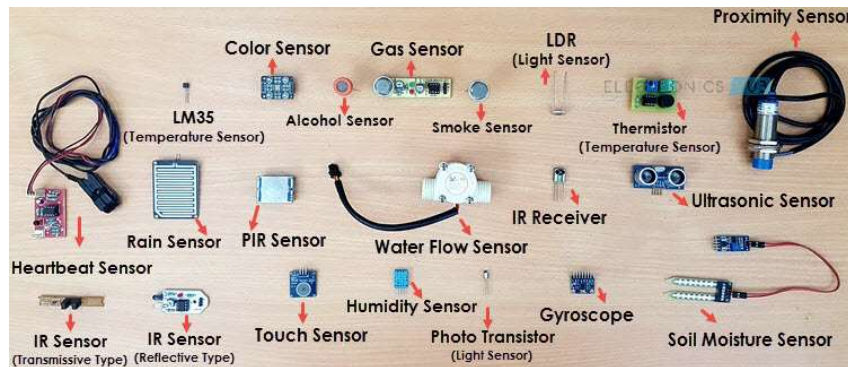
What is Agent?

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What is a Sensor?



What is a Sensor?

There are numerous definitions as to what a sensor is but I would like to define

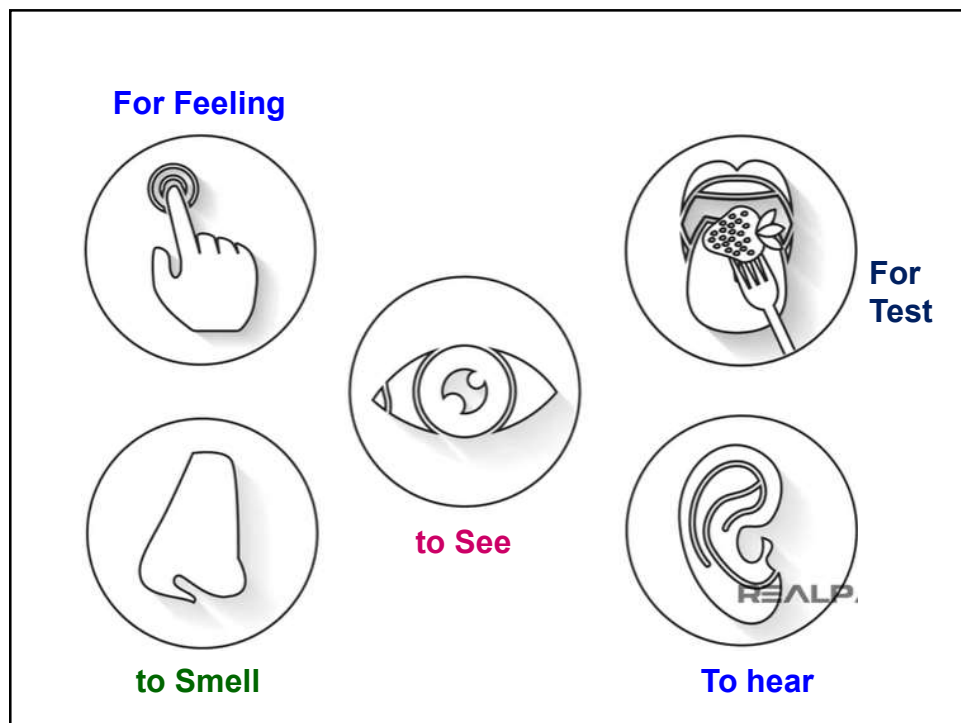
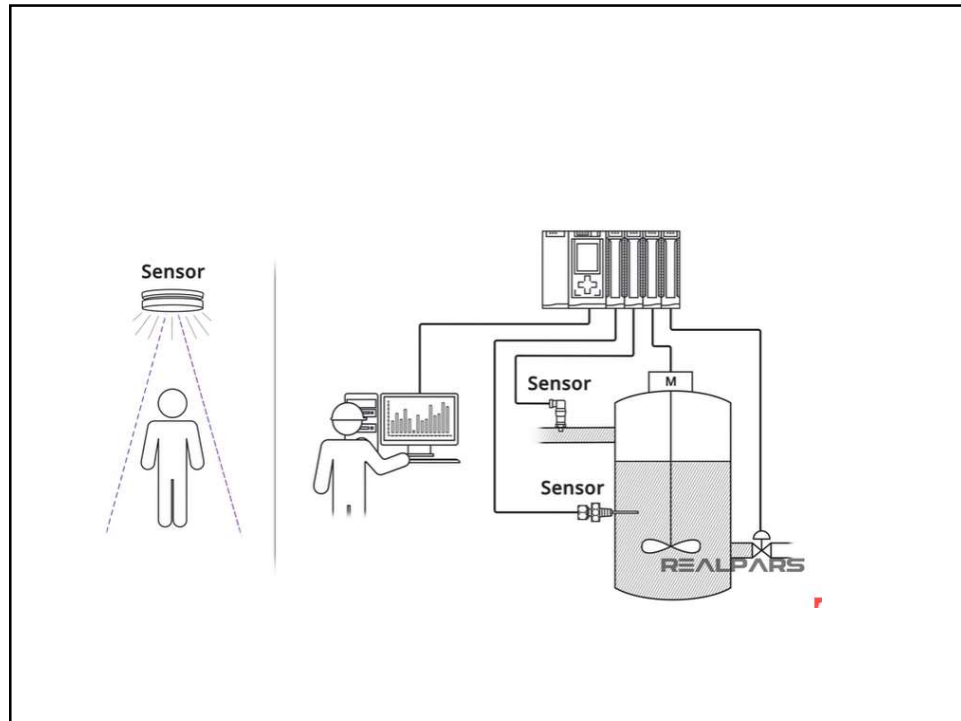
A Sensor as an input device which provides an output (signal) with respect to a specific physical quantity (input).

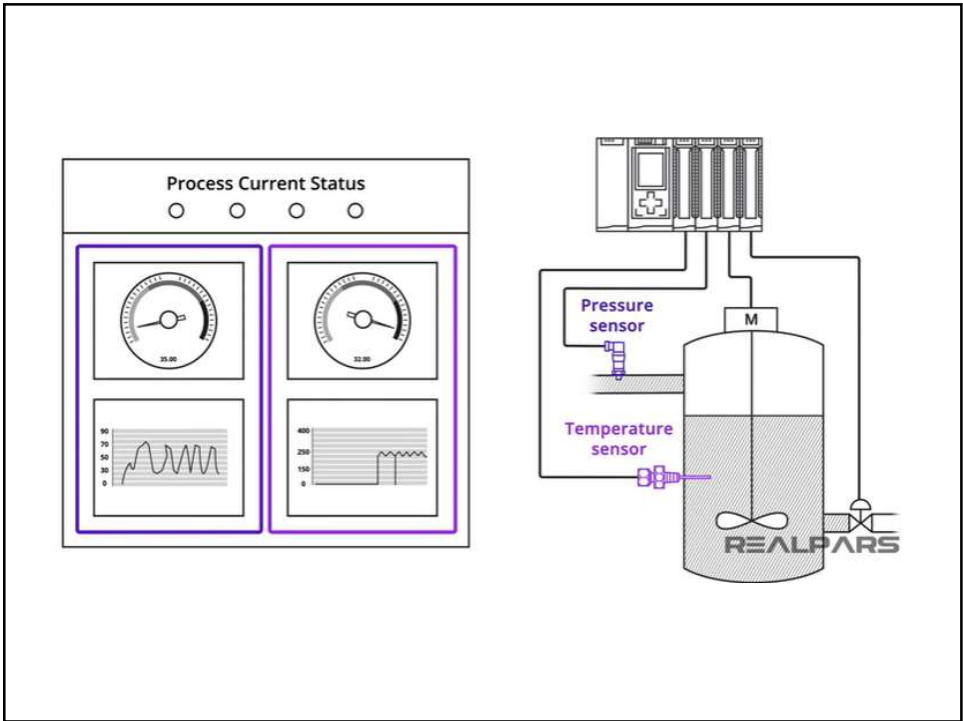
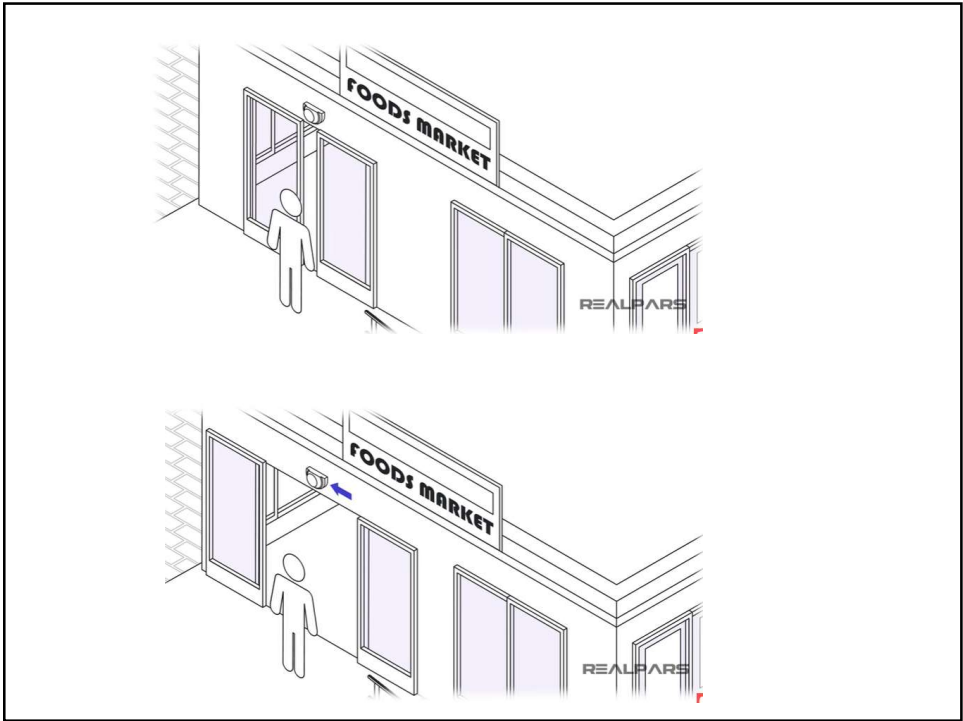
The simplest example of a sensor is an LDR or a Light Dependent Resistor.

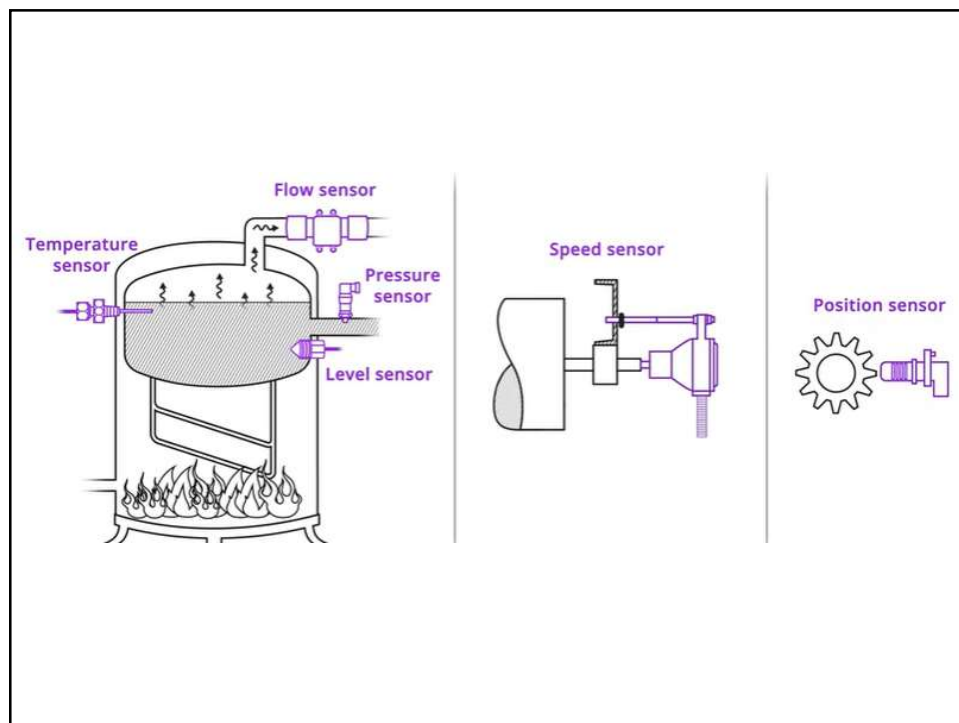
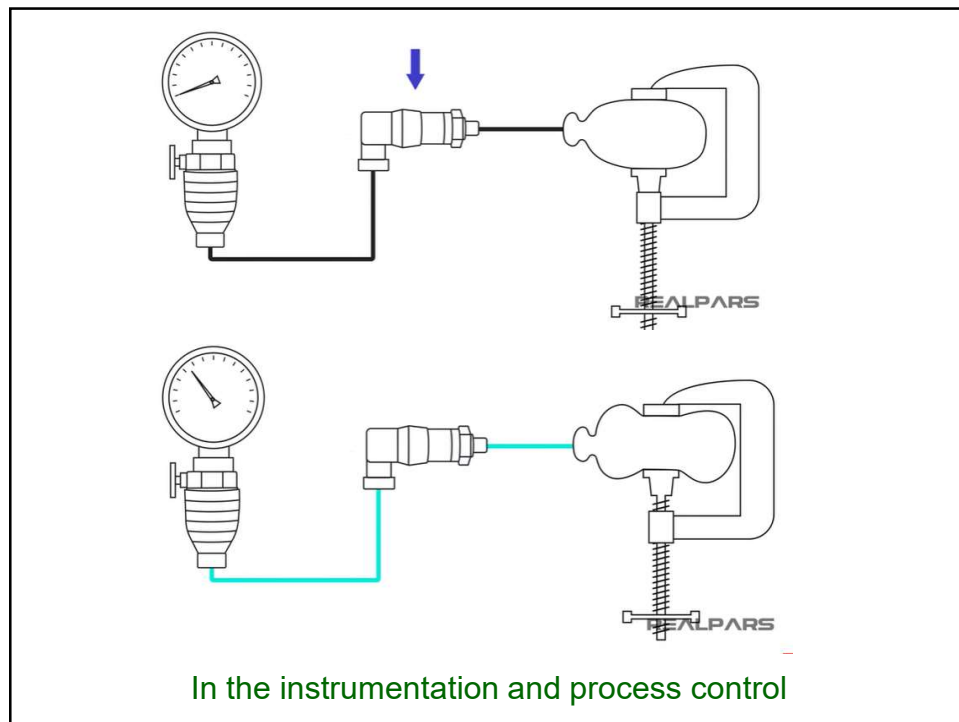
It is a device, whose resistance varies according to intensity of light it is subjected to. When the light falling on an LDR is more, its resistance becomes very less and when the light is less, well, the resistance of the LDR becomes very high.

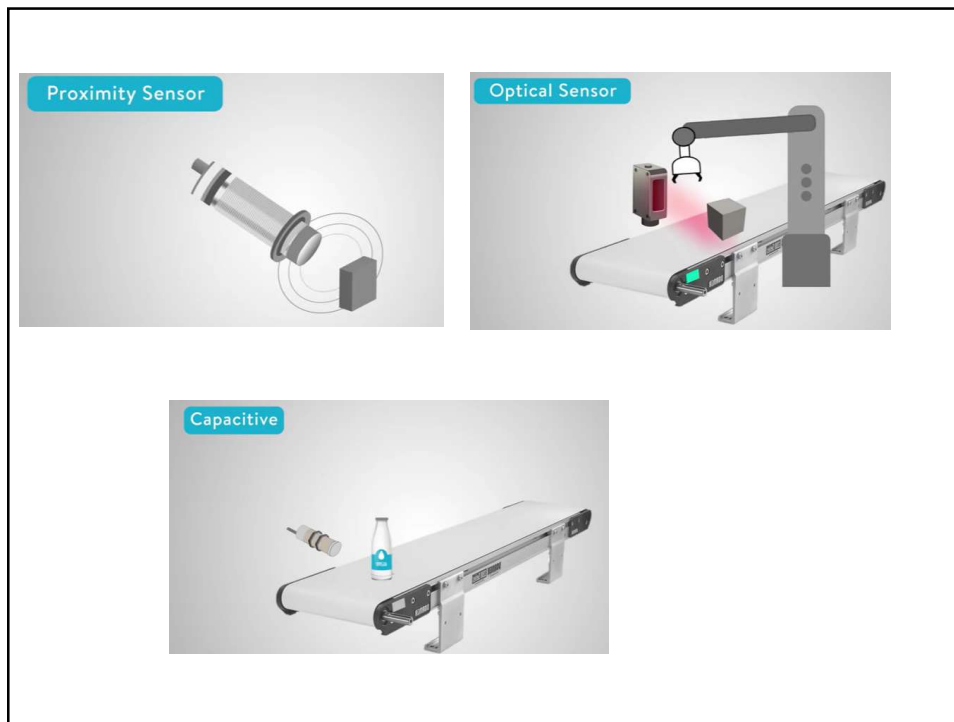
We can connect this LDR in a voltage divider (along with other resistor) and check the voltage drop across the LDR. This voltage can be calibrated to the amount of light falling on the LDR. Hence, a Light Sensor.

Now that we have seen what a sensor is, we will proceed further with the classification of Sensors.









Based on input the sensors are classified as

(i) **Active** and (ii) **Passive**

Active Sensor:

An active sensor is a device that emits its own energy (such as light, radio waves, or sound waves) toward an object or environment and then measures the energy that is reflected or returned. It requires an external excitation signal or a power signal.

•Example: Radar, LiDAR, sonar.

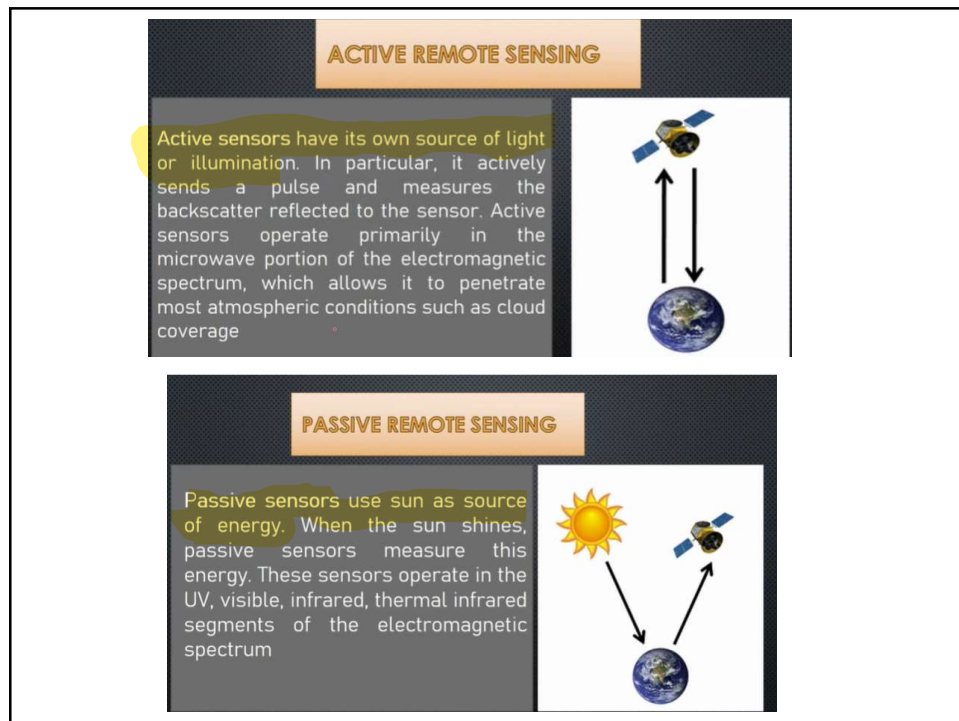
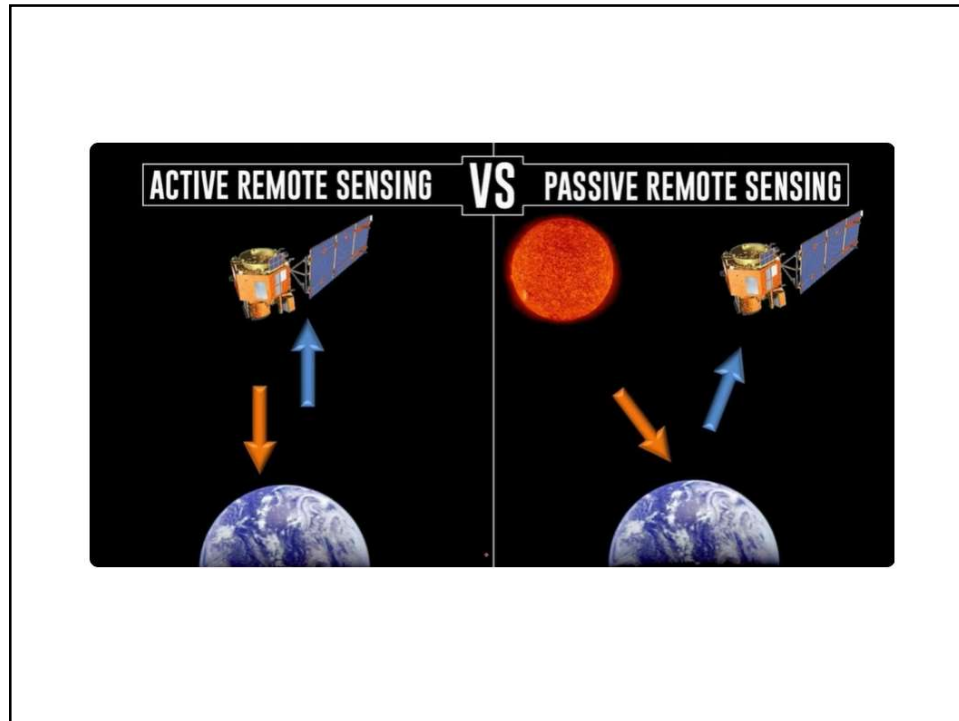
•Key point: It provides its own source of illumination or signal.

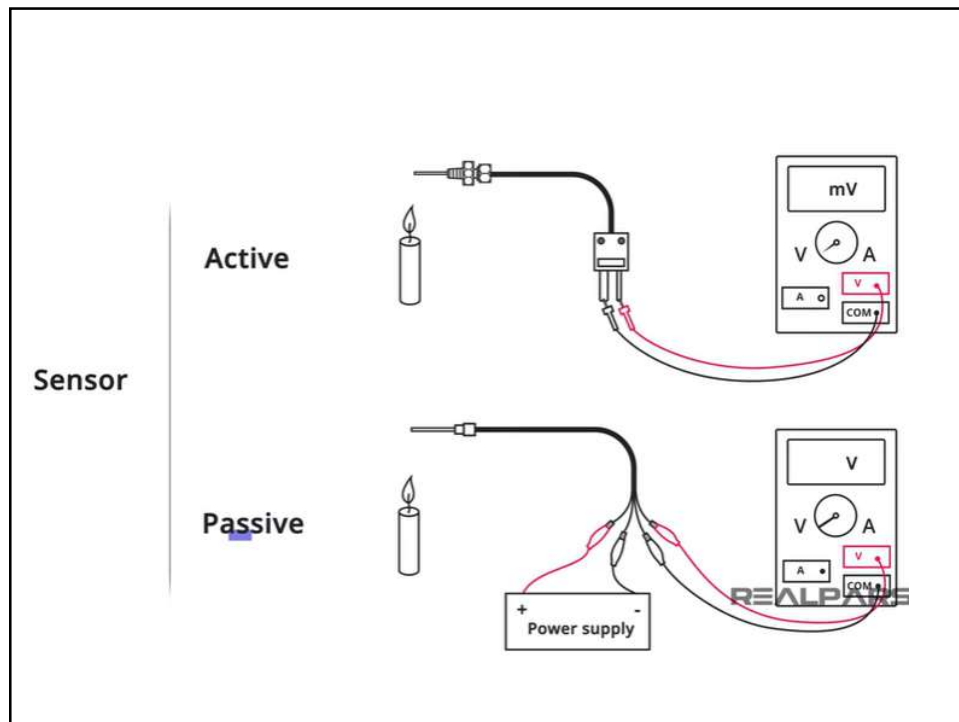
Passive Sensor:

A passive sensor is a device that does not emit energy of its own, but instead detects natural energy (such as sunlight, heat, or radiation) that is reflected or emitted by objects. It does not require any external power signal and directly generates output response.

•Example: Cameras, infrared sensors, radiometers, human eyes.

•Key point: It relies on external sources of energy (like the Sun or thermal radiation).





Based on Means of Detection:

The other type of classification is based on the means of detection used in the sensor. Some of the means of detection are Electric, Biological, Chemical, Radioactive etc.

Based on Conversion Phenomenon:

The next classification is based on conversion phenomenon i.e., the input and the output. Some of the common conversion phenomena are Photoelectric, Thermoelectric, Electrochemical, Electromagnetic, Thermo-optic, etc.

The final classification of the sensors are

(i) Analog and (ii) Digital

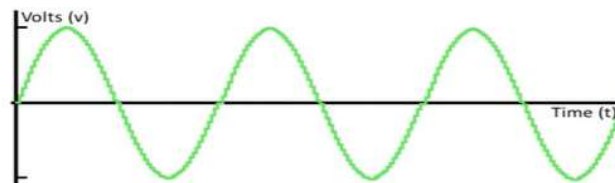
Analog Sensors produce an analog output i.e., a continuous output signal (usually voltage but sometimes other quantities like Resistance etc.) with respect to the quantity being measured.

Digital Sensors, in contrast to Analog Sensors, work with discrete or digital data. The data in digital sensors, which is used for conversion and transmission, is digital in nature

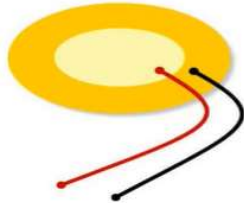
Analog Sensors?

Analog sensors connect through the analog inputs. These Sensors produce a continuous output signal which is generally proportional to the quantity being measured. Physical quantities such as Temperature, Speed, Pressure, Displacement, Strain etc are all analog as they tend to be continuous in nature.

Example -A potentiometer attached to the door hinge. This is an analog input. It will allow you to detect the amount that the door is open or closed. For example, you could use it to tell if the door was 50% open, 12.3376...% open, or any other amount between completely closed (0% open) and completely open (100% open).



List of some analog Sensors?



Pressure sensor will produce an analog output signal that is proportional to the amount of applied pressure.



Sensors that detect changes in position, velocity, orientation, shock, vibration, and tilt by sensing motion are called as accelerometers.

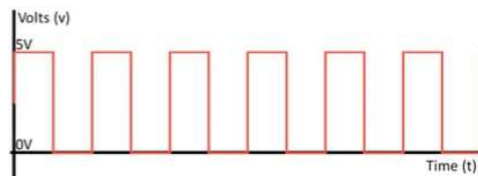


Sensors that are used for detecting the amount of light striking the sensors are called as light sensors.

Digital Sensors?

Devices which perform an "Input" function are commonly called **Sensors** because they "sense" a physical change in some characteristic that changes in response to some excitation, for example heat or force and convert that into an electrical signal.

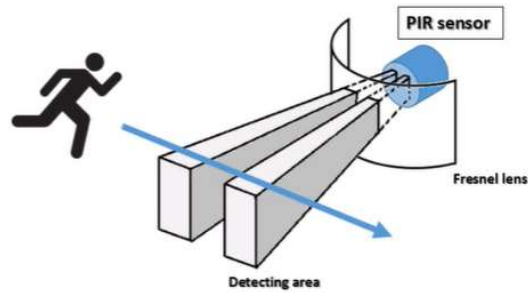
Example-An on/off switch placed on the door frame. This is a digital input. It will allow you to detect when the door is closed and when it is open, but will not detect anything between these states



Digital Sensors



PIR Sensor-detects the infrared radiations coming from the human body in the surrounding area.



Different Types of Sensors

The following is a list of different types of sensors that are commonly used in various applications. All these sensors are used for measuring one of the physical properties like Temperature, Resistance, Capacitance, Conduction, Heat Transfer etc.

- 1. Temperature Sensor**
- 2. Proximity Sensor**
- 3. Accelerometer**
- 4. IR Sensor (Infrared Sensor)**
- 5. Pressure Sensor**
- 6. Light Sensor**
- 7. Ultrasonic Sensor**
- 8. Smoke, Gas and Alcohol
Sensor**
- 9. Touch Sensor**
- 10. Color Sensor**

11. Humidity Sensor
12. Position Sensor
13. Magnetic Sensor (Hall Effect Sensor)
14. Microphone (Sound Sensor)
15. Tilt Sensor
16. Flow and Level Sensor
17. PIR Sensor (Pyroelectric Sensor)
18. Touch Sensor
19. Strain and Weight Sensor

Proximity Sensor: A proximity sensor detects the presence or absence of an object without making physical contact. It works by emitting an electromagnetic field or a beam of radiation (infrared, ultrasonic, etc.) and observing the changes.

Example: In smartphones, the proximity sensor turns off the touchscreen when you hold the phone close to your ear during a call to prevent accidental touches.

Accelerometer: An accelerometer measures acceleration forces, which can be static (like gravity) or dynamic (caused by movement or vibration). It is often used to detect orientation, tilt, and motion.

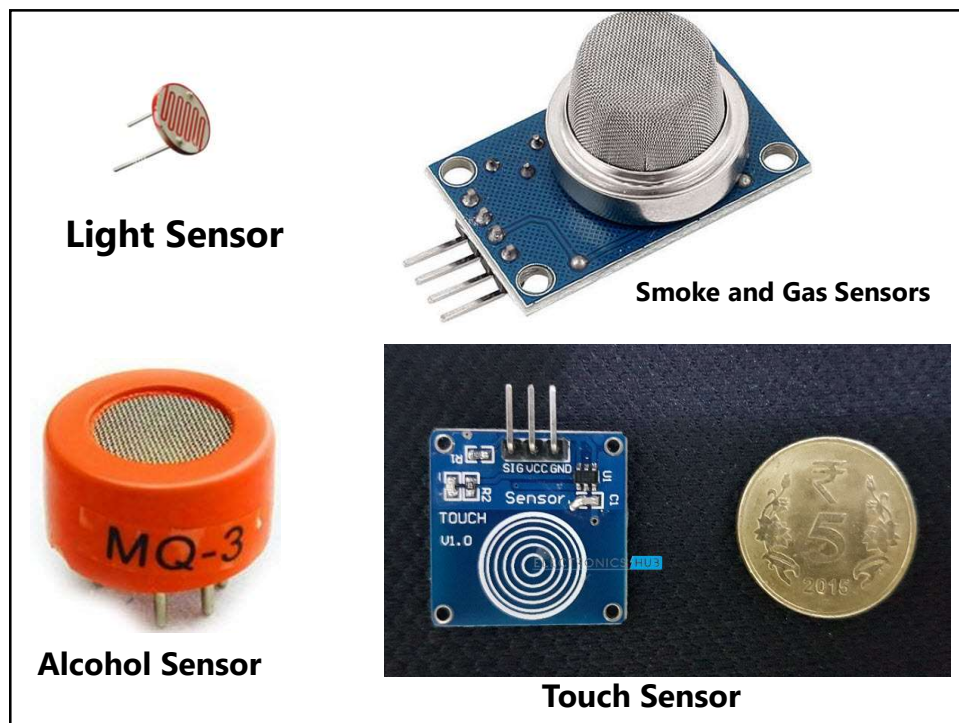
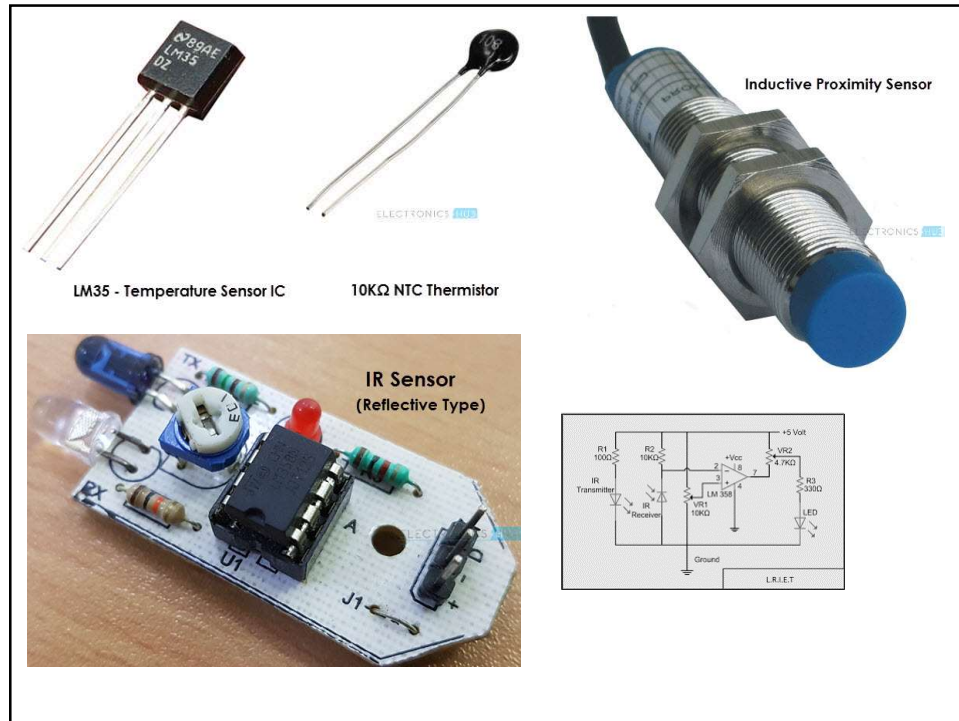
Example: In mobile phones, the accelerometer changes the screen orientation automatically when you rotate the phone (portrait ↔ landscape).

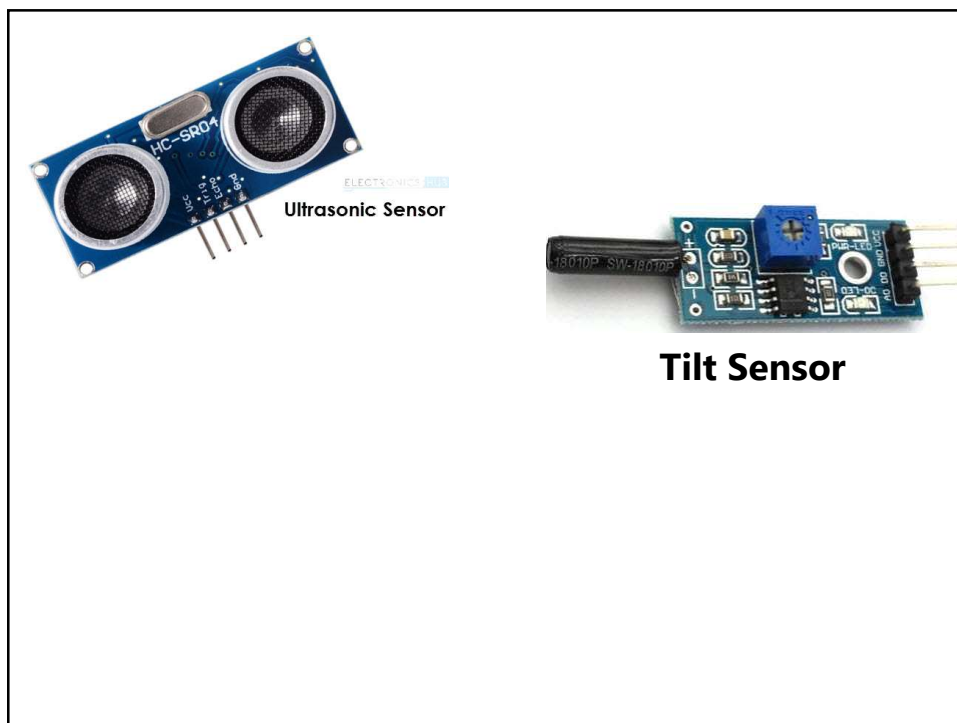
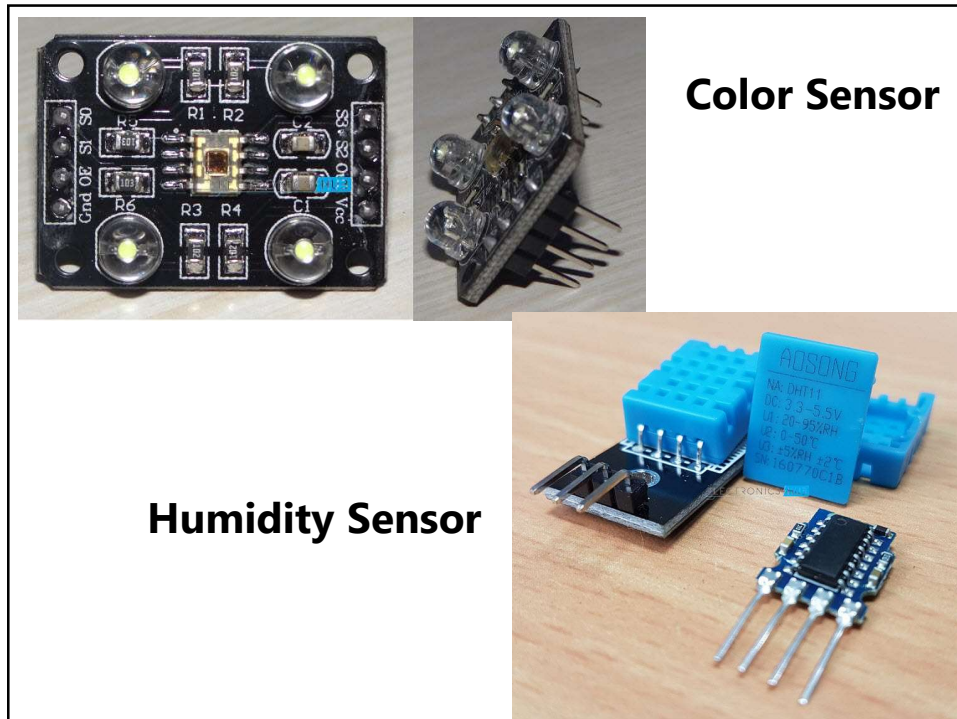
IR Sensor (Infrared Sensor): An infrared (IR) sensor detects infrared light (heat radiation) from objects in its surroundings. It can be used for object detection, distance measurement, and thermal sensing.

Example: The remote control of a TV uses an IR sensor in the TV to receive infrared signals sent from the remote.

Ultrasonic Sensor: An ultrasonic sensor measures distance by sending out ultrasonic sound waves and calculating the time taken for the echo to return after bouncing off an object.

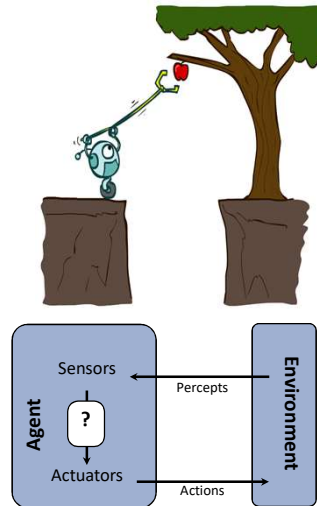
Example: In cars, ultrasonic sensors are used in reverse parking systems to detect nearby obstacles and alert the driver with beeps.



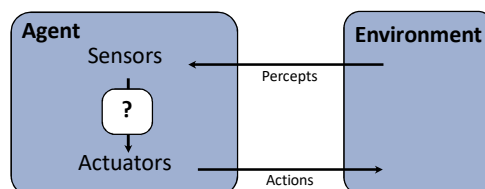
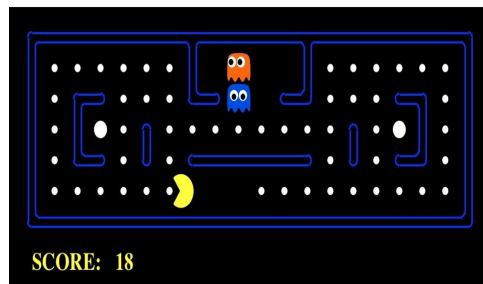


Designing Rational Agents

- An **agent** is an entity that *perceives* and *acts*.
- A **rational agent** selects actions that maximize its (expected) **utility**.
- Characteristics of the **percepts**, **environment**, and **action space** dictate techniques for selecting rational actions



Pac-Man as an Agent



Pac-Man is a registered trademark of Namco-Bandai Games, used here for educational purposes

Demo1: [pacman-l1.mp4](#) or L1D2

Agent/ Artificial Agent

Goals of Agent

- High performance
- Optimized result
- Rational (right) Action



A robotic agent has?

A robotic agent:

- cameras and infrared range finders for sensors;
- various motors for actuators

A software agent has?

A software agent:

- keystrokes, file contents, received network packages as sensors
- displaying on the screen, writing files, sent network packets as actuators

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Agent

Is a human agent?

A human agent has?

- A human agent:
 - eyes, ears, nose, tongue, skin, and other organs for sensors;
 - hands, legs, mouth, and other body parts for actuators or effectors.

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Agent

A human agent has eyes, ears, and other organs for sensors and hands, legs, vocal tract, and so on for actuators.

A robotic agent might have cameras and infrared range finders for sensors and various motors for actuators.

A software agent receives keystrokes, file contents, and network packets as sensory inputs and acts on the environment by displaying on the screen, writing files, and sending network packets.

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Agent Terminology/Agent-Related Terms

Performance Measure of Agent:
It is the criteria, which determines how successful an agent is.

Behavior of Agent:
It is the action that agent performs after any given sequence of percepts.

Percept:
It is agent's perceptual inputs at a given instance.

Percept Sequence:
It is the history of all that an agent has perceived till date.
Or
A complete history of everything the agent has ever perceived.

Agent Function (or Policy):
It is a map from the precept sequence to an action.

Agent Program:
Implements the agent function.

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Table and agent program

We can imagine **tabulating the agent function** that describes any given agent; for most agents, this would be a very large table—infinite, in fact, unless we place a bound on the length of percept sequences we want to consider. Given an agent to experiment with, we can, in principle, construct this table by trying out all possible percept sequences and recording which actions the agent does in response.

The **table** is, of course, an external characterization of the agent.

Internally, the agent function for an artificial agent will be implemented by an **agent program**. It is important to keep these two ideas distinct. The agent function is an abstract mathematical description; the agent program is a concrete implementation, running within some physical system.

Simple Example vacuum-cleaner world

To illustrate these ideas, we use a very simple example—the vacuum-cleaner world shown in Figure 2.2. This world is so simple that we can describe everything that happens; it's also a made-up world, so we can invent many variations. This particular world has just two locations: squares A and B. The vacuum agent perceives which square it is in and whether there is dirt in the square. It can choose to move left, move right, suck up the dirt, or do nothing. One very simple agent function is the following: if the current square is dirty, then suck; otherwise, move to the other square. A partial tabulation of this agent function is shown in Figure 2.3 and an agent program that implements it appears in Figure 2.8 on page 48.

Vacuum-cleaner world



To illustrate these ideas, we use a very simple example—the **vacuum-cleaner world** shown in Figure below.

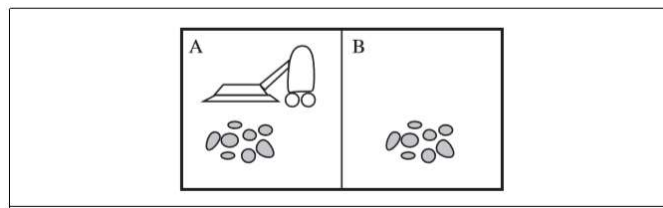
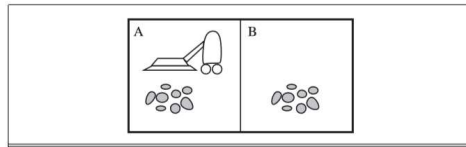


Figure 2. A vacuum-cleaner world with just two locations

This world is so simple that we can describe everything that happens; it's also a made-up world, so we can invent many variations. This particular world has just two locations: squares A and B.

Vacuum-cleaner world



The vacuum **agent perceives** which square it is in and whether there is dirt in the square. **It can choose to**

move left
move right
suck up the dirt
or do nothing.

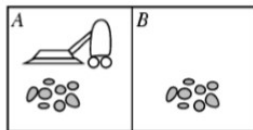
One very simple agent function is:

If the current square is dirty, then suck; otherwise, move to the other square.

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Vacuum-cleaner world

A **partial tabulation** of this agent function is shown in Figure 3.



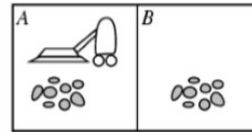
- Two locations: A and B
- Percepts: location and contents, e.g., [A,Dirty]
- Actions: *Left, Right, Suck, NoOp*

Percept Sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck

Figure 3. Partial tabulation of a simple agent function for the vacuum-cleaner world shown in above figure.

Vacuum-cleaner world

A **partial tabulation** of this agent function is shown in Figure 3.



Percept Sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean],[A, Clean]	Right
[A, Clean],[A, Dirty]	Suck
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
⋮	⋮

Figure 3. Partial tabulation of a simple agent function for the vacuum-cleaner world shown in above figure.

Vacuum-cleaner world

Looking at Figure 3, we see that various vacuum-world agents can be defined simply by filling in the right-hand column in various ways.

The obvious question, then, is this: What is the right way to fill out the table? In other words, what makes an agent good or bad, intelligent or stupid? We answer these questions in the next section.

An **agent program** that implements it appears in Figure 4.

```

function REFLEX-VACUUM-AGENT([location,status]) returns an action
  if status = Dirty then return Suck
  else if location = A then return Right
  else if location = B then return Left
  
```

Figure 4. The agent program for a simple reflex agent in the two-state vacuum environment.

This program implements the agent function tabulated in Figure 3.

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Ideal Rational Agent

What is Ideal Rational Agent?

An **ideal rational agent** is the one, **which is capable of doing expected actions to maximize its performance measure**, on the basis of its perceive and the actions it can perform.

A rational agent is one that does the right thing.

What does it mean to do the right thing?

A rational agent always performs right action,
where the right action means the action that causes the agent to be most successful.

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Rational Agent

Rational Agent

A rational agent is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.

Rationality maximizes expected performance, while perfection maximizes actual performance

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Rational Agent

Rational Agent

This leads to a definition of a rational agent:

For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

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Is this (Vacuum-cleaner) Rational Agent?

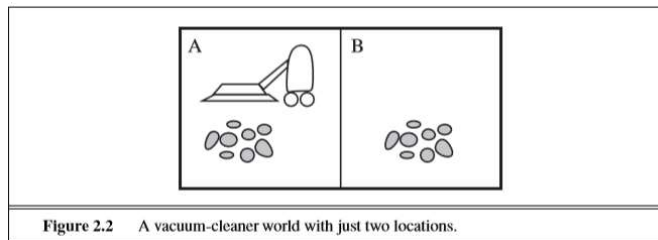


Figure 2.2 A vacuum-cleaner world with just two locations.

Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
⋮	⋮

Figure 2.3 Partial tabulation of a simple agent function for the vacuum-cleaner world shown in Figure 2.2.

Is this (Vacuum-cleaner) Rational Agent?

Let us assume the following :

- The performance measure awards one point for each clean square at each time step, over a “lifetime” of 1000 time steps
- The “geography” of the environment is known *a priori* (Figure 2.2) but the dirt distribution and the initial location of the agent are not. Clean squares stay clean and sucking cleans the current square. The *Left* and *Right* actions move the agent left and right except when this would take the agent outside the environment, in which case the agent remains where it is.
- The only available actions are ***Left***, ***Right***, and ***Suck***
- The agent correctly perceives its location and whether that location contains dirt.

Is this (Vacuum-cleaner) Rational Agent?

Under these circumstances, we claim that the agent is indeed rational.

One can see easily that the same agent would be irrational under different circumstances. For example, once all the dirt is cleaned up, the agent will oscillate needlessly back and forth; if the performance measure includes a penalty of one point for each movement left or right, the agent will fare poorly. A better agent for this case would do nothing once it is sure that all the squares are clean. If clean squares can become dirty again, the agent should occasionally check and re-clean them if needed. If the geography of the environment is unknown, the agent will need to explore it rather than stick to squares *A* and *B*. Exercise 2.2 asks you to design agents for these cases.



Rational Agent

Rationality of an agent depends on the following four factors :

- The **performance measures** that determine the degree of success.
- The agent's **prior knowledge of the environment**.
- The **actions** that the agent can perform
- Agent's **percept sequence** to date.

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In our discussion of the rationality of the simple vacuum-cleaner agent, we had to specify the performance measure, the environment, and the agent's actuators and sensors.

It depends upon:

- (i) What the **P**erformance measure
- (ii) What is known about the **E**nvironment
- (iii) What **A**ctuators the agent has
- (iv) What **S**ensors the agent has



Task Environment and PEAS

We group all these under the heading of the task environment.

For the acronymically minded, we call this the **PEAS** (Performance, Environment, Actuators, Sensors) description.

In designing an agent, **the first step must always be to specify the task environment as fully as possible.**

The problem the agent solves is characterized by (PEAS)

PEAS:

- Performance Measure,
- Environment,
- Actuators, and
- Sensors.

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Ideal Rational Agent

Performance measure is an objective criterion for success of an agent's behavior

Performance measure:

- e.g., performance measure of a vacuum-cleaner agent could be
 - amount of dirt cleaned up,
 - amount of time taken,
 - amount of electricity consumed,
 - amount of noise generated, etc.

A more suitable measure would reward the agent for having a clean floor

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Agent

PEAS for an automated taxi driver

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Figure 2.4 PEAS description of the task environment for an automated taxi.

- **Performance measure:** Safe, fast, legal, comfortable trip, maximize profits
- **Environment:** Roads, other traffic, pedestrians, customers
- **Actuators:** Steering wheel, accelerator, brake, signal, horn
- **Sensors:** Cameras, sonar, speedometer, GPS, odometer, engine sensors, keyboard

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PEAS for an automated taxi driver

First, what is the performance measure to which we would like our automated driver to aspire? Desirable qualities include getting to the correct destination; minimizing fuel consumption and wear and tear; minimizing the trip time or cost; minimizing violations of traffic laws and disturbances to other drivers; maximizing safety and passenger comfort; maximizing profits. Obviously, some of these goals conflict, so tradeoffs will be required. Next, what is the driving environment that the taxi will face? Any taxi driver must deal with a variety of roads, ranging from rural lanes and urban alleys to 12-lane freeways. The roads contain other traffic, pedestrians, stray animals, road works, police cars, puddles, and potholes.

The taxi must also interact with potential and actual passengers. There are also some optional choices. The taxi might need to operate in Southern California, where snow is seldom a problem, or in Alaska, where it seldom is not. It could always be driving on the right, or we might want it to be flexible enough to drive on the left when in Britain or Japan.

Obviously, the more restricted the environment, the easier the design problem. The actuators for an automated taxi include those available to a human driver: control over the engine through the accelerator and control over steering and braking. In addition, it will need output to a display screen or voice synthesizer to talk back to the passengers, and perhaps some way to communicate with other vehicles, politely or otherwise. The basic sensors for the taxi will include one or more controllable video cameras so that it can see the road; it might augment these with infrared or sonar sensors to detect distance to other cars and obstacles. To avoid speeding tickets, the taxi should have a speedometer, and to control the vehicle properly, especially on curves, it should have an accelerometer. To determine the mechanical state of the vehicle, it will need the usual array of engine, fuel, and electrical system sensors. Like many human drivers, it might want a global positioning system (GPS) so that it doesn't get lost. Finally, it will need a keyboard or microphone for the passenger to request a destination.

In Figure 2.5, we have sketched the basic PEAS elements for a number of additional agent types.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Figure 2.5 Examples of agent types and their PEAS descriptions.



Agent

PEAS for a medical diagnosis system

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Agent

PEAS for a medical diagnosis system

- **Performance measure:** Healthy patient, minimize costs, lawsuits
- **Environment:** Patient, hospital, staff
- **Actuators:** Screen display (questions, tests, diagnoses, treatments, referrals)
- **Sensors:** Keyboard (entry of symptoms, findings, patient's answers)

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Agent

PEAS for a satellite image analysis system

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Agent

PEAS for a satellite image analysis system

- **Performance measure:** correct image categorization
- **Environment:** downlink from orbiting satellite
- **Actuators:** display categorization of scene
- **Sensors:** color pixel arrays

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Agent

PEAS for a part-picking robot

- **Performance measure:** Percentage of parts in correct bins
- **Environment:** Conveyor belt with parts, bins
- **Actuators:** Jointed arm and hand
- **Sensors:** Camera, joint angle sensors

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Agent

PEAS for Interactive English tutor

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Agent

PEAS for Interactive English tutor

- **Performance measure:** Maximize student's score on test
- **Environment:** Set of students
- **Actuators:** Screen display (exercises, suggestions, corrections)
- **Sensors:** Keyboard

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Agent

PEAS for a refinery controller

- **Performance measure:** maximize purity, yield, safety
- **Environment:** refinery, operators
- **Actuators:** valves, pumps, heaters, displays
- **Sensors:** temperature, pressure, chemical sensors

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Structure of Agent



Table:

We can imagine **tabulating** the agent function **that describes any given agent**.

This would be a very large table—infinite, in fact, unless we place a bound on the length of percept sequences we want to consider.

The table is an external characterization of the agent.

- Simplest agent program is a table-driven agent

```
function TABLE-DRIVEN-AGENT(percept) returns an action
  static: percepts, a sequence, initially empty
          table, a table of actions, indexed by percept sequences, initially
              fully specific
  append percept to the end of percepts
  action ← LOOKUP(percepts, table)
  return action
```

This is a BIG table...clearly not feasible!

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Thank You...